

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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20 September 2026

Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence* (collective phase order), *compatibility* (local descriptions agree where their domains overlap) and *self-representation* (the state carries an accurate reconstruction of itself). Coherence supplies neither of the others, and five results establish that with no cortical hypothesis. Synchrony bounds content disagreement only through encoders shared on overlaps, and then only at the patch the bound is read on, not over the whole population. The resultant does not determine the state: a winding field and a uniform one are stationary at one coupling, at opposite ends of that observable. A unique fixed point is not a self-model, since a contraction can discard its input: accuracy across a family of states requires distinguishable codes. Where the encoding is a region's restriction, that resolution is demanded of the region before any readout is chosen. And agreement about a change takes communication. Four of the five are checkable for a machine candidate, which fixes what a claim of machine unity or self-representation commits its maker to measuring. Under stated hypotheses, compatible

local states then glue to a unique global state that an accurate contracting reconstruction fixes uniquely. Installed energy bounds the mean coupling a declared actuator supplies, so a coherent stationary branch requires it to exceed $2D/\kappa$; κ is derived here for neither cortex nor a processor. A Lean 4 development carries these results and their composition, with the eight connecting hypotheses as explicit arguments, no incomplete proofs, and an audited axiom footprint no wider than its libraries'. Further controls bound what observation can decide: distinct mechanisms share stationary phase distributions, and coherent adaptation can lose the structure it was meant to acquire. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, which a within-subject emergence-from-anaesthesia design can test; the finite-size estimate here is biased below 1/2 by its escape criterion, as a deterministic reference confirms. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling. Testing it requires calibrated geometry, resolved phase dynamics and validated decoders.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase

11 order. *Compatibility* concerns whether local descriptions agree where their
12 domains overlap. *Self-representation* concerns whether the resulting state
13 carries an accurate internal reconstruction of itself. We propose the latter
14 two relations as formal correlates of conscious unity and a minimal self; their
15 interpretation requires evidence beyond the phase model.

16 Unity has been approached differently elsewhere: integrated information
17 theory treats integration and exclusion as constitutive (Tononi et al., 2016),
18 global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene
19 et al., 2014), and predictive-processing accounts derive both from a hierar-
20 chical generative model (Friston, 2010). Here the three requirements are
21 kept apart rather than derived from one another, so that each carries its own
22 assumptions and admits its own measurement; Section 9.4 returns to the
23 comparison.

24 Consider reaching for a cup. A visual description of its location and a
25 description of the hand’s relation to it can agree about where the cup is while
26 contributing different contents. Synchronizing the two processes would leave
27 disagreement about that shared quantity unresolved. An internal representa-
28 tion must additionally locate the perceiver within the represented situation.
29 This example motivates the requirements; the mathematical constructions
30 use spatial measures and fields in real normed vector spaces, whose identifi-
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these
33 requirements and the assumptions each connection needs. A declared ac-

34 tuator’s stored energy bounds its mean coupling. Phase encoders shared on
35 overlaps convert phase order into a bound on content disagreement. Compat-
36 ible spatial measures have a unique global extension; approximate agreement
37 controls a specified global selection. A contracting reconstruction makes that
38 extension its unique fixed point only if it accurately reconstructs the exten-
39 sion. Accuracy across a family of states imposes a further condition on the
40 information retained by the encoding.

41 The empirical contribution is to identify what observations can decide.
42 Stationary phase distributions can be shared by stronger coupling, weaker
43 diffusion and a matched common drive. Finite ramps change the observable
44 onset. A winding field is stationary at the same coupling as a uniform one and
45 has the opposite resultant, so that observable does not determine the state.
46 Coherent adaptation can lose alignment with a designated environmental
47 template, and a compatibility estimator needs diagnostics that distinguish
48 content from decoder artifacts. These findings determine the measurements
49 needed to test the proposed connections.

50 Three of the ingredients are standard: the synchronization threshold
51 $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing
52 of compatible local sections, and the Banach fixed-point theorem (Banach,
53 1922). Four things here are not. The installed-energy condition bounds
54 mean coupling by the stored energy of a declared actuator, which makes a
55 resource condition necessary for coherence on the stationary branch. The
56 coherence-to-content bound converts phase order into a statement about de-

57 descriptions rather than about timing, and its patch-local form is what keeps
58 it from being vacuous in a large population. A stability criterion on the
59 reweighted coupling kernel, together with an existence condition on an am-
60 plitude layer, settles when a low resultant reports weak coupling and when
61 it reports a winding field. And the composition is machine-checked together
62 with a witness discharging all eight hypotheses at once, which is what makes
63 their conjunction non-empty rather than a list no object satisfies.

64 Granting those premises, collected in Table 1, the account runs as follows.
65 Extracellular contraction on awakening raises endogenous field amplitude
66 and with it the effective coupling between cortical oscillators. Once that
67 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
68 within a quarter turn the population settles onto it; other initial data leave
69 windings stationary at the same coupling. Encoders shared where cortical
70 territories overlap turn that phase order into agreement about the quantities
71 the territories describe in common, so the local descriptions glue: there is
72 one state of which each is a restriction. Part of that state encodes the state
73 itself, and a readout of that part reconstructs it accurately enough, and
74 sensitively enough across the states that matter, to make it the unique state
75 the system models itself as being. On this account the glued state is what
76 makes distributed activity one situation, and its accurate self-reconstruction
77 is what gives that situation a perspective. Every step of it is conditional.

78 Section 2 collects the results that hold whatever physical field is proposed
79 and whatever substrate carries it; a reader who rejects the cortical hypothesis

80 of Section 8 still has to accommodate them. Sections 3–5 then develop the
81 three requirements together (Fig. 1). Section 6 composes them, stating the
82 thermodynamic constraints and all eight connecting hypotheses per branch;
83 its supporting models of feedback, learning and resource supply have distinct
84 state spaces and assumptions. Section 7 then examines what the observables
85 can establish, before Section 8 develops the cortical hypothesis and an awak-
86 ening experiment. The hypothesis is that extracellular contraction increases
87 field-mediated coupling; its direction, rate and relation to recovery require
88 independent calibration. Supplemental Material organizes the constructions
89 and controls behind the claims they support.

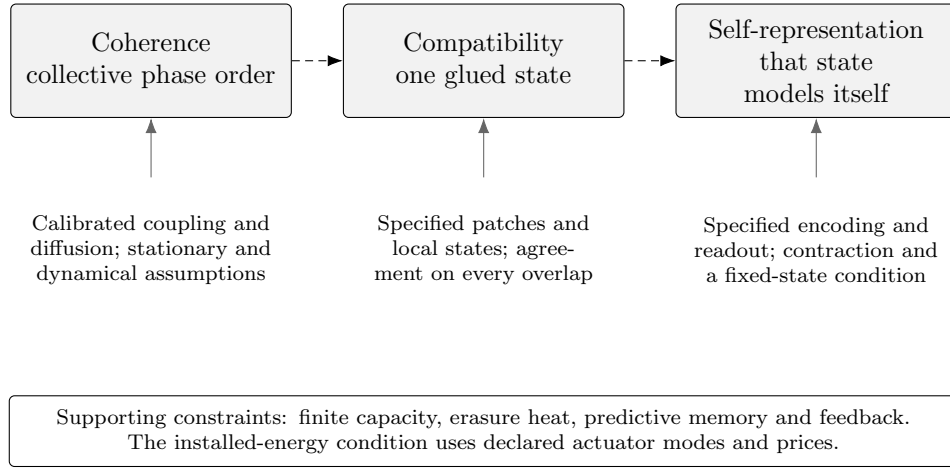


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

90 2 What holds without the cortical hypothesis

91 Most of what follows is conditional. Table 1 names eight hypotheses that
92 the composition consumes rather than derives, and the cortical identification
93 of Section 8 is a further commitment on top of them. Five results are not
94 conditional in that way. They constrain any account that reads unity off
95 phase coherence, whatever physical field is proposed to carry the coupling
96 and whatever substrate carries the field. Each is stated below where it is
97 used; collected here, they are the article’s fixed points.

98 **Phase order alone does not constrain content.** Assign every patch
99 the same phase and let two local mass profiles disagree at a shared site.
100 The profiles are chosen independently of phase, so synchrony bounds their
101 disagreement not at all, and they have no common extension: perfect locking
102 is consistent with local descriptions that do not glue (Section 4). What the
103 positive result of Section 4.2 costs is exactly this gap. A bound on content
104 disagreement needs encoders already related on overlaps, and that relation
105 is a supplied fact about the descriptions rather than a consequence of the
106 dynamics.

107 **The order parameter does not determine the state.** A field winding
108 once across a periodic sheet has $r = 0$ while every patch of it stays locked,
109 and this is not a contrived initial condition. On a periodic array of any
110 dimension, a winding is stationary for *any* isotropic symmetric kernel at

111 identical frequencies, because the coupling term is odd in the separation
 112 against a kernel even in it (`char_is_kuramoto_trajectory`). Section 7.3
 113 exhibits a sheet on which uniform and winding initial conditions reach 0.9657
 114 and 0.0083 at one coupling, diffusion and kernel. A low resultant therefore
 115 reports either weak coupling or a winding field, and no measurement in r
 116 alone separates them. The state also carries an integer the resultant does
 117 not: read around a ring, the same winding has winding number q , and a
 118 nonzero q obstructs every global real-valued phase lift (`winding_degree_`
 119 `obstructs`). The resultant is an average over sites and moves when they are
 120 reweighted or the patches redrawn; the degree telescopes and does not.

121 **A unique fixed point is not a self-model.** A constant readout con-
 122 tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness
 123 is consequently the wrong criterion, and Section 5.2 gives the right shape
 124 of one: reconstructing M states pairwise more than 2ε apart within error ε
 125 requires M distinguishable codes, so an alphabet of 2^b codes caps the family
 126 at $M \leq 2^b$. This is a counting argument and holds for any encoder and
 127 readout whatever.

128 **Where the encoding is a restriction, the region must resolve the**
 129 **family.** Read the code as the state's own restriction to a region and
 130 the criterion moves off the readout entirely. A boundary whose region
 131 reads two relevant states identically cannot reconstruct them more than 2ε
 132 apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant

133 boundary that does reconstruct a separated family restricts injectively on
134 it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the
135 region sits is fixed before an encoding is chosen, so no readout supplies
136 resolution the region lacks.

137 **Agreement after a change takes communication.** In a synchronous
138 network whose updates read only the site’s own state and its declared neigh-
139 bours’ messages, two configurations agreeing on a site’s causal past through
140 T rounds leave that site identical at round T . Two interventions outside
141 that past that require reports more than 2ε apart therefore cannot both be
142 reported from it within ε by the deadline (Section 5). Snapshot compatibil-
143 ity and timely compatibility are therefore different requirements, and cutting
144 the path removes the second while leaving the first intact.

145 None of these depends on the coupling being electromagnetic, on the sub-
146 strate being cortex, or on the eight hypotheses of Table 1 holding. Section 9.3
147 applies four of them — all but the one about the resultant, which is a state-
148 ment about a spatial observable — to a candidate that is neither biological
149 nor field-coupled.

150 **3 Coherence: physical coupling and its re-** 151 **source condition**

152 **3.1 Dynamics and the stationary threshold**

153 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

154 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
155 K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field
156 coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct
157 parameters. The stationary calculation below uses identical frequencies and
158 this scalar coupling. Applying it to a spatial kernel requires an aggregation
159 rule and a check of the resulting approximation.

160 For identical frequencies in their rotating frame, the positive normalized
161 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

162 where I_n are modified Bessel functions and ψ is the collective phase. The for-
163 mal derivation uses the absence of stationary current for a periodic gradient
164 drift to obtain this density, then verifies normalization and self-consistency.
165 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above

threshold, exactly one positive solution coexists with zero and increases
strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first
harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
exactly above threshold. About the coherent density, the quadratic form
is strictly dissipative transverse to collective rotation, whose rate is zero.
These linear estimates concern the limiting equation. Relating them to finite
stochastic trajectories requires a dynamical mean-field limit and branch selec-
tion; exchange symmetry alone does not establish the needed independence
(Sznitman, 1991).

3.2 Installed energy bounds mean coupling

A spatial model replaces the finite sum by integration against a kernel. One
declared implementation carries response modes with continuous profiles ϕ_i ,
installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

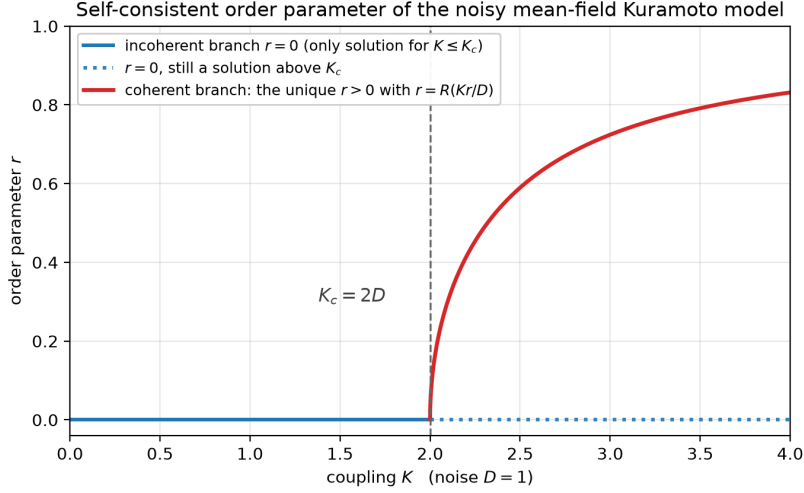


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

181 The same configuration sets the reciprocal kernel and stored energy. An
 182 executed transition changes the kernel through its changed occupancies and
 183 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 184 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 185 pair reconstruction on refining spatial partitions converges to its product-
 186 measure integral under the specified compactness and sampling assumptions;
 187 the refinement index is independent of the number of control updates.

188 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 189 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

190 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_z p(z)U(z), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned}
 \tag{5}$$

191 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 192 coherent stationary branch and any growing Fourier mode about the uniform
 193 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 194 is necessary: energy above it need not produce sufficient coupling because
 195 the bound can be loose.

196 The quantity constrained is installed energy rather than dissipated heat,
 197 and how much the condition excludes depends entirely on κ , which the mode
 198 profiles, prices and substrate normalization supply: any macroscopic sub-
 199 strate carries installed energy far above the thermal floor, so the inequality
 200 binds through that conversion and not through the energy, and no result
 201 here derives κ for cortex or for any other candidate. What the condition
 202 contributes is a resource requirement in units a measurement can reach. Ap-
 203 plying the scalar threshold to heterogeneous or signed spatial dynamics re-
 204 quires a justified reduction, and identifying the installed kernel with a cortical
 205 field is a separate physical commitment.

206 4 Compatibility: from coordinated activity 207 to a global description

208 4.1 Why common timing is insufficient

209 For overlapping patches U and V , compatibility means that restricting either
210 local description to $U \cap V$ gives the same object. Descriptions can agree on
211 shared quantities while contributing different information elsewhere. Com-
212 patible sections of the specified sheaf have a unique global extension. On
213 a finite cover, local finite spatial measures agreeing as measures on every
214 overlap also have a unique finite-measure extension whose sheaf image is the
215 glued section. No mass-one normalization is imposed.

216 In the cup example, visual and reaching descriptions must express a
217 shared quantity, such as cup location, in common coordinates. Restriction
218 retains that quantity while dropping details particular to each description.
219 Agreement permits surface appearance and hand configuration to contribute
220 to one situation. The mathematical counterexample assigns the same phase
221 to every patch while two local mass profiles disagree on a shared site. Those
222 profiles have no common extension. Because they are chosen independently
223 of phase, synchrony places no constraint on their disagreement.

224 This distinction limits the biological interpretation. Receptive territories
225 and content domains must be related to spatial patches, and the descriptions
226 must support the required extension. Matching overlap marginals of de-

227 coded probability laws need not determine a unique global joint distribution
 228 (Abramsky & Brandenburger, 2011).

229 The cover is itself a choice, and the construction constrains it in one way
 230 only: synchronized patches agree where they overlap. Where they do not
 231 overlap there is nothing to compare, so a cover by pairwise disjoint patches
 232 satisfies that obligation for *every* assignment of local descriptions and phases
 233 (`ofDisjointCover`) and the gluing theorem still returns a unique global sec-
 234 tion, which is the tuple it was handed (`glue_unique_of_disjoint`). Member-
 235 ship of the class then reports nothing about the system beyond the patches
 236 covering it. What a gluing claim asks of a cover is therefore a non-degeneracy
 237 the theorem does not assume: that distinct patches meet (`HasNonempty-`
 238 `Overlaps`), and that what they share is a quantity fixed independently of
 239 the agreement it is being used to establish.

240 4.2 When coherence does constrain content

241 A declared relation between phase and content gives a positive result. Let
 242 contents take values in a real normed vector space E . Patch i reads an
 243 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 244 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 245 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 246 sites; their equality on overlaps is a supplied relation between descriptions of
 247 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 248 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every

249 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1-r^2}. \quad (6)$$

250 Here discrepancy is the norm of the difference, and r and N are read on
251 whichever population the bound is applied to. That choice is what decides
252 whether the estimate says anything. A patch carries an order parameter of
253 its own, and reading the bound there carries that patch's resultant and that
254 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`
255 `coherence`), so a large population observed through small patches is con-
256 strained by the small count. Read instead on the whole array, the factor N
257 makes the estimate weak, because it draws a pointwise guarantee from an
258 average over N^2 pairs. At $r = 1$ the observations agree exactly at either
259 scale. A shared scalar encoder is a special case. Dropping the overlap rela-
260 tion permits disagreement even at perfect locking. A common nonexpansive
261 linear projection preserves agreement, but agreement of a projection does
262 not establish agreement of the full vector.

263 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
264 constrains anything only where $\sqrt{2} N\sqrt{1-r^2}$ falls below that. The winding
265 sweep of Section 7.3 fixes both ends of that range on one sheet. Read on the
266 sweep's own patch, 625 sites at patch-local order 0.9015, the bound is $382.5 L$
267 and constrains nothing; holding that order fixed, the count alone brings it
268 below the maximum chord only at 3 sites or fewer. A nearest-neighbour pair

269 of the same winding, the smallest patch there is, gives $0.1388 L$. Shrinking
 270 patches therefore thins the overlaps at the same rate as it tightens the bound,
 271 and the two ends are not reached by the same cover. At a columnar grain
 272 the estimate constrains agreement inside one locally synchronous population
 273 and is vacuous across the distant territories whose agreement unity is about.

274 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 275 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 276 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 277 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

278 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 279 duce an initial mismatch. Approximate observations leave a nondecaying
 280 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 281 update to the phase process.

282 4.3 What approximate agreement determines

283 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 284 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 285 Weights that vanish outside their patches and sum to one at each site select
 286 a global profile by weighted averaging. The selection lies within ε of each
 287 patch there, and any two such selections differ by at most ε , independently
 288 of cover multiplicity. An L -Lipschitz readout consequently varies by at most

289 $L\varepsilon$ across selections.

290 At zero tolerance, the selection is the unique exact extension. At every
291 positive tolerance there are families with no exact extension; selection then
292 gives an approximation whose dependence on weights is bounded. Supple-
293 mental Material gives a three-site witness attaining the bound. The mass-
294 profile construction uses the scalar case of the content dynamics and addi-
295 tionally requires nonnegativity of the initial profiles and observations. The
296 general vector result supplies no identification with the spatial-measure sheaf.

297 These constructions concern declared objects: a specified cover, contents
298 valued in a normed space, and encoders related on overlaps. They select
299 neither the neural cover nor the decoder, and identifying their contents with
300 perceptual content, or establishing that the identification bears on experi-
301 ence, is a further modelling task; Section 7 sets out what a measurement of
302 it would have to establish.

303 **5 Self-representation: a state that models it-** 304 **self**

305 **5.1 The reflexive construction**

306 Compatibility gives a global state. Reflexivity relates it to an encoding within
307 that state. Equip global sections with a complete metric. If the additional
308 restriction-resonance condition identifies encoding with restriction to a region

309 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (8)$$

310 The region and readout are fixed across the state space. If F contracts and
 311 fixes the section s_* formed from the local descriptions, Banach's theorem
 312 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
 313 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
 314 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
 315 one for $\tau > 0$. Constructing a map with that factor remains a modelling
 316 obligation.

317 In the cup example, the global state includes the represented scene and the
 318 perceiver's relation to it. A local encoding must retain enough information
 319 for the readout to recover that modelled state; cup position alone would be
 320 insufficient if hand configuration and viewpoint could not be reconstructed.

321 For the same map with Lipschitz factor $q < 1$, approximate reconstruction
 322 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (9)$$

323 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 324 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 325 weakens.

326 5.2 Reconstruction across a family of states

327 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 328 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 329 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

330 Thus M relevant states pairwise more than 2ε apart require at least M
 331 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 332 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 333 existing self-prediction map.

334 This is an additional accuracy criterion on a family, with no requirement
 335 for multiple exact fixed points. A blind encoder or constant readout fails it
 336 for separated states despite having a unique fixed point.

337 Where the encoding is read as the state's own restriction to a region, the
 338 criterion becomes a condition on the region. Restriction resonance requires
 339 the encoder and that restriction to identify the same pairs of states, so a
 340 boundary reconstructing two relevant states more than 2ε apart while its re-
 341 gion reads them identically is not resonant (`not_isRestrictionResonance_`
 342 `of_reconstructs`); read forwards, a resonant boundary that reconstructs
 343 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
 344 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
 345 cally is fixed by where the region sits, before an encoding is chosen, so the

346 resolution the family demands is the region's and no readout supplies it. The
 347 implication runs one way: a region that does resolve the family is not thereby
 348 shown to carry a resonant encoding, and E89 supplies no such resolution.

349 Those two demands are jointly satisfiable with nothing folded anywhere.
 350 Take the region to be the whole substrate: restriction is then the identity, so
 351 the boundary is resonant, reconstructs every declarable family exactly, and
 352 restricts injectively on all of them (`topAvatar_satisfies_obligations`). A
 353 properness condition on the region (`IsProperAvatar`) is therefore a separate
 354 requirement rather than a consequence of resonance and resolution, and what
 355 a fold into a local region is asked for. Its quantitative form replaces the
 356 alphabet by a metric. With an L -Lipschitz readout, a family reconstructed
 357 within ε whose members lie pairwise more than r apart has δ -separated codes
 358 whenever $\delta L \leq r - 2\varepsilon$, so the family is no larger than the δ -packing number
 359 of what the region carries (`encard_le_packingNumber_restrictToAvatar`).
 360 That is the shape a continuous family of states needs, and at the whole
 361 substrate it says nothing, because the region's range is everything. The gain
 362 L is declared and derived here for no physical readout, and at unbounded
 363 gain the bound says nothing.

364 **5.3 Agreement after a change takes communication**

365 Snapshot compatibility does not specify how quickly descriptions respond
 366 when a shared quantity changes. In a synchronous network whose updates
 367 read only the site's own state and declared neighbours' messages, two initial

368 configurations agreeing on a site’s causal past through T rounds produce the
369 same state there at round T . The deadline consequence is the one Section 2
370 states, and it follows from the reconstruction bound in Section 5.2: a report
371 from that site is a readout of a code that the two interventions leave identical.

372 A three-site delay line witnesses failure after one round and exact report-
373 ing after two; cutting the path prevents delivery at every round. A report
374 may still happen to be correct about one fixed world. The bound concerns
375 responsiveness across interventions.

376 Throughout this section *prediction* names a reconstruction map, and the
377 relevant family of states, the metric, the tolerance, the communication graph
378 and the deadline are all declared inputs, as are the represented variables and
379 the fidelity counted as accurate. Reading the map as temporal neural pre-
380 diction would require dynamics; reading the resulting code count as physical
381 memory capacity would require a hardware model, which a state-dependent
382 readout would have to extend.

383 6 Thermodynamics and composition

384 The phase and representation results concern coupling, content disagreement
385 and reconstruction error. Thermodynamics constrains the physical opera-
386 tions that could implement them, and it does so from the side rather than
387 along the spine: no edge below derives coherence, compatibility or reconstruc-
388 tion from a heat bound, and the resource condition these edges terminate in

389 is necessary without being restrictive for any macroscopic candidate (Sec-
 390 tion 3.2). What this half contributes is a ledger in which a resource claim
 391 is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$;
 392 erasure heat depends on specified system–bath assumptions (Landauer, 1961;
 393 Norton, 2005; Shenker, 2000). A separate boundary result concerns a field
 394 reaching disconnected vacuum components (Kibble, 1976). The composition
 395 connects these objects through the explicit hypotheses in Table 1.

396 **6.1 Feedback changes the information constraint**

397 For passive memory X_t driven by an environmental signal S_t , the predictive
 398 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

399 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
 400 et al., 2012). Zero retained information also satisfies it, so a task and a
 401 learning rule require additional specification. Relating this bound to Lan-
 402 dauer heat requires a common thermal scale and budget, and memory waste
 403 bounded by erased entropy.

404 An acting agent changes the signal it will receive. Specify policy, world,
 405 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned} \quad (12)$$

406 The second conditional term vanishes in the passive model. With feedback,
 407 the signed difference can be negative as action creates correlations with the
 408 future. A causal reading requires an environmental state sufficient to describe
 409 the relevant dynamics.

410 For a controlled substep, the active thermodynamic bound instead uses
 411 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 412 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
 413 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
 414 2013). Every forward path of positive probability must have a positive-
 415 probability reverse for this finite identity. A missing reverse gives infinite
 416 divergence. The supplement develops the support conditions and composes
 417 actuation with memory updates on their shared intermediate law.

418 **6.2 Learning and repeated operation need explicit re-** 419 **sources**

420 The finite examples distinguish selecting a feasible policy, relaxing a register
 421 toward an encoded reward, and learning an unknown task from observa-
 422 tions. The observational learner acquires correlation with an environmental
 423 parameter while its action marginal stays uniform. A continuing model exe-
 424 cutes actuation, observation-driven learning and reset in sequence, preserving
 425 above-chance performance at a positive work cost per cycle.

426 A phase-derived sensor makes the informational distinction concrete: an

427 informative readout and a blind control have the same phase order but dif-
 428 ferent learning performance. Attaching the learner’s register to the actuator
 429 of Section 3.2 ties reward, installed kernel and work to its own executed law.
 430 The learning witness rises above the necessary installed-energy bound while
 431 its matched control remains on the exclusion line. These finite constructions,
 432 detailed in Supplemental Material, leave the evolving feedback between phase
 433 dynamics, installation and sensing to be specified.

434 For that model, common energy E , thermal scale θ , local detailed balance
 435 and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

436 where expected cumulative work stays within the supplied allowance B at
 437 every prefix. This bounds expectations. Separate constructions track a store
 438 on each trajectory, include finite sources, charge sensor clearing and prepend
 439 preparation. They specify when a mean allowance represents an available
 440 resource; an expected budget alone permits overdrawn paths.

441 These constructions establish costs for their declared processes. A finite
 442 source with a positive net cost permits only finitely many funded cycles,
 443 while continuing operation requires replenishment. Register–bath witnesses
 444 also distinguish actual bath-energy transfer from transition-log heat. Iden-
 445 tifying a work store, an erasure budget and the installation energy of Sec-
 446 tion 3.2 requires physical relations that these separate models do not supply

447 automatically.

448 6.3 What the composition establishes

449 The passive theorem `chain` and feedback theorem `chain_active` each ac-
 450 cept eight connecting implications (Table 1). In the active branch, a named
 451 substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

452 with B the register’s heat. The upper allocation follows from a ledger whose
 453 operations deliver that total heat to one reservoir and whose other operations
 454 do not increase joint entropy. Those ledger assumptions are physical inputs.
 455 Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.

Table 1 (continued)

Connection	Status	Required assumption
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register's thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations' compressiveness.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours.
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.

Table 1 (continued)

Connection	Status	Required assumption
Installed energy	Conditional theorem	Under Section 3.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above it. E12 is equivalent to its conclusion, so capacity and boundary are independent roots rather than a step from one to the other; both E45 discharges take their convergence from refinement data and not from the thermodynamic bound; E78 states that coherence relaxes the supplied cover, and the coherence-to-content estimate of Eq. (6) bounds disagreement only for encoders already shared on overlaps. The inferential content of the composition is its last step, where contraction and a fixed glued section give uniqueness. What composing them establishes is that the premises are consistent, named in one place and machine-checked together, and not that coherence follows from installed energy or self-representation from coherence.

Joint witnesses combine a register, boundary, refining mesh and small

474 spatial cover; active witnesses add controlled thermal updates and a priced
 475 installation. They establish satisfiability under their stated identifications.
 476 Mesh convergence follows from refinement data, and the witnesses establish
 477 no common cortical mechanism. Supplemental Material maps the claims to
 478 their Lean declarations and records the counterexamples. The development
 479 declares no additional axioms beyond its libraries' foundational axioms.

480 **7 What the observations establish**

481 The composition holds under its premises. Whether observation can estab-
 482 lish those premises is a separate question, and the three requirements it
 483 connects concern different objects: phase dynamics, local descriptions and
 484 internal encodings. Their observables must retain these distinctions. The
 485 following controls show which mechanisms remain indistinguishable, which
 486 temporal approximations fail, which states one observable cannot separate,
 487 and what independent content measurements must establish. Each is a finite
 488 numerical regime on a declared model, with full parameters and controls in
 489 Supplemental Material, and none of them calibrates a cortical quantity.

490 **7.1 Stationary distributions leave the mechanism un-** 491 **determined**

492 Equation (2) gives a concentration-coherence relation with no fitted shape
 493 parameter. Estimate r from circular resultant length and a from density

494 shape: inverting R from the measured r would make agreement automatic.
 495 Even independently estimated agreement identifies a statistical family shared
 496 by different mechanisms.

497 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
 498 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
 499 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
 500 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
 501 and diffusion D_0 have that same density. Their stationary distributions co-
 502 incide window by window despite different causes.

503 Resolved dynamics supply additional information. In the ideal model with
 504 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
 505 increment means constrain drift. Separating endogenous coupling from a
 506 matched common drive additionally requires a measured drive or controlled
 507 perturbation; identifying electromagnetic coupling requires evidence specific
 508 to that interaction. Supplemental Material gives the equivalence construction
 509 and its numerical check.

510 **7.2 Finite ramps change the observable onset**

511 The square-root law in Eq. (3) describes stationary solutions. A changing
 512 coupling requires time for the phase distribution to approach those solutions.
 513 In a deterministic mean-field control that crosses the threshold symmetrically,
 514 none of the scanned rates meets the declared tracking tolerance: slowing
 515 the crossing does not remove the terminal residual under that initialization.

516 A slow control variable therefore supplies no general guarantee of branch
517 tracking at onset. Supplemental Material gives the residual, rate range and
518 admissible rates away from threshold.

519 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
520 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
521 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
522 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
523 6 uncensored legs supply the delay scaling below; the decade-spaced ones
524 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
525 onset exponents differ from those of the exact stationary branch evaluated
526 on the same grids. At the fastest speed the ensemble mean never reaches the
527 fit window, so its fitted slope does not estimate stationary onset.

528 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
529 makes the apparent exponent depend on ramp speed and fitting window.
530 The specified deviation-based critical ramp speed is not bracketed by this
531 sweep. With fluctuating coupling, the relevant stationary comparison also
532 depends on rate: the mean coupling approximates order in the tested fast
533 regime, while averaging stationary order over coupling works for the tested
534 slow drives confined to one side of threshold. Slow crossing drives can satisfy
535 neither approximation.

536 **The prediction.** The delay scaling is what this section supports as a pre-
537 diction, and it needs no absolute calibration. Over the 6 uncensored legs the

mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of $[0.394, 0.494]$, which does not contain $1/2$. What that shortfall measures is the escape criterion and not the scaling. Applied to a deterministic mean-field run seeded at the fluctuation floor, the same estimator returns 0.425 over the same legs at the same escape criterion, inside the interval on the measurement. That run has no noise to delay it, so the shortfall is the criterion's: it crosses the saturation between the floor it starts from and the level it is stopped at, and tightening the criterion to $r \geq 0.05$ returns 0.490. The bias is reproduced where the answer is known, which is what licenses reading the measurement as the law rather than as a competing exponent.

Stated as a law rather than as a number: onset delay scales as the square root of the rate at which coupling crosses threshold. Testing it requires only that $\dot{K}$ be monotonic in a manipulated rate — not its value, nor κ , nor γ — so it is a within-subject design that varies the rate of emergence from anaesthesia alone, with each participant as their own control. Such a design must avoid the hazard this sweep exhibits: a fixed absolute escape criterion meets a size-dependent critical fluctuation floor, so the criterion has to be referred to each recording's own pre-threshold fluctuation scale rather than to a shared number. The diffusion D is the remaining input, and Section 7.1 notes that it is estimable from resolved phase traces through the rate $2D$ at which quadratic variation accumulates, which is a different quantity from the concentration the propofol exercise could not range over.

Additional controls address integration bias, population size and spatial

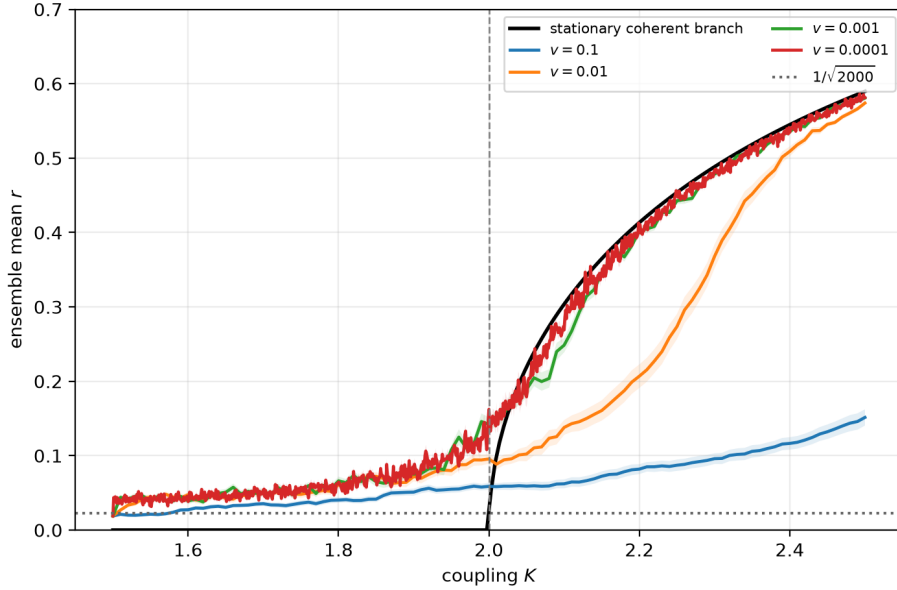


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

561 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
562 $[1.958, 2.009]$ contains 2.

563 **7.3 A winding state leaves the order parameter am-** 564 **biguous**

565 The order parameter r is indifferent to where a phase sits in space. A field
566 winding once across a periodic sheet therefore has $r = 0$ while every patch
567 of it stays locked. Such a field is not an artificial construction here. On a

568 torus a winding is stationary for any isotropic symmetric kernel at identical
 569 frequencies: the coupling term is odd in the separation against a kernel even
 570 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves
 571 this for any finite periodic array, this sheet included. A winding satisfies the
 572 positivity and identical-frequency hypotheses. It violates only the quarter-
 573 turn initial spread that the convergence result additionally requires before
 574 placing the limit at the uniform state.

575 On the sheet of Section 3's spatial control, a winding survives below
 576 an interpolated decay length of 0.1871 mm and unwinds above it (Fig. 4).
 577 That boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same
 578 sheet at the same coupling, diffusion and kernel reaches $r = 0.9657$ from a
 579 uniform start and $r = 0.0083$ from a winding one. Both are stationary, with
 580 patch-local order 0.9015 and no winding defect. At fixed K the observable
 581 therefore takes two values spanning nearly its whole range. Coupling does
 582 not determine the resultant, and a low resultant does not by itself distinguish
 583 weak coupling from a winding field.

584 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-
 585 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 586 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
 587 Heterogeneity shortens the range that sustains a wave without removing it.

588 Which windings persist has a criterion behind it. Linearizing the phase
 589 model about a winding returns a coupling of the same translation-invariant
 590 form, each separation reweighted by the cosine of the phase the winding

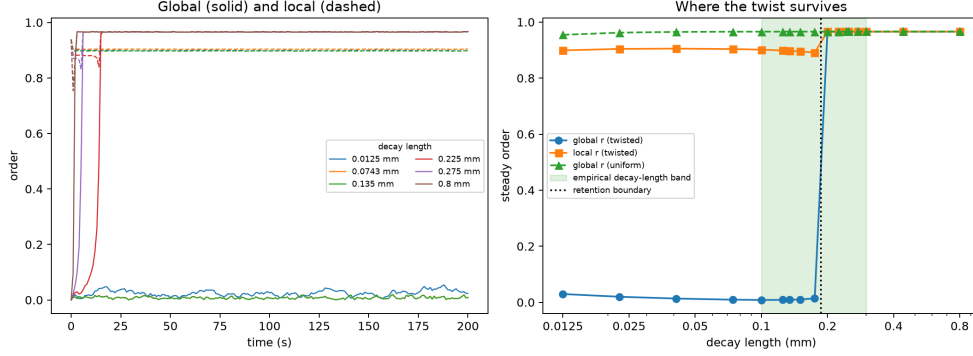


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

591 advances across it (`hasDerivAt.charJacobian`). That reweighted kernel is
 592 again even in the separation, so the same characters diagonalize it and ev-
 593 ery mode's rate is one of its eigenvalues. Where the winding advances by
 594 at most a quarter turn across every separation the coupling reaches, the
 595 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 596 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 597 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 598 longer separations is what carries a phase advance past a quarter turn, and
 599 that is the mechanism the decay-length sweep varies. The criterion bounds
 600 rates rather than trajectories: asymptotic stability is not proved, so the

601 states that persist occupy a narrower range than it admits, and the bound-
602 ary quoted above is measured and not derived from it. The same kernel is the
603 potential's curvature, so a winding at $r = 0$ is not the global minimum the
604 convergence result places its limit at, and one carrying a growing mode is not
605 a local minimum either: that result's quarter-turn hypothesis must exclude
606 a winding rather than merely decline to cover it (`char_not_potential_min`).

607 An amplitude layer decides which of these states exist, and leaves what
608 the resultant reports unchanged. Every oscillator of the phase model sits on
609 the unit circle, so a field with an amplitude node has no representation in
610 those dynamics. The real Stuart–Landau extension supplies one on the same
611 coupling, and leaves intact the gradient structure the convergence results rest
612 on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i = (\mu - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its
613 phase equation is the Kuramoto equation with each coupling weighted by a
614 ratio of amplitudes (`amplitude_phase_is_kuramoto`). The phase model is
615 this one with the amplitude ratios set to one, rather than a separate object.

616 What the added degree of freedom decides is which windings exist at all.
617 On a periodic array of any dimension, this sheet included, a winding requires
618 $\mu + \lambda > 0$. Here $\lambda = \sum_d f_d(\cos \phi_d - 1)$ is the eigenvalue the kernel supplies,
619 summed over separations, with ϕ_d the phase the winding advances across d
620 (`exists_charState_iff`; on the ring $\phi_d = 2\pi qd/n$). For the nonnegative
621 kernels used here $\lambda \leq 0$, and moving coupling mass to longer separations
622 lowers it further. A broad enough kernel therefore removes the winding out-
623 right. The sweep above reaches that same boundary by a different route:

624 the phase model retains every winding at every coupling, and loses this one
625 under integration instead. The two layers bound the same regime from op-
626 posite sides. A phase recording does not separate a state the amplitude layer
627 excludes from one the dynamics fails to sustain.

628 Three limits bound the layer for a reader of the cortical literature. These
629 states are windings and not spirals, and on the sheet that is a theorem rather
630 than a shortage of dimensions: equal modulus at every site puts the phase
631 circulation at exactly zero on every plaquette, and on a periodic sheet the
632 circulations of *any* field sum to zero, so a lone defect is not a state of this
633 geometry at all. Relaxational dynamics has no propagation speed, which is
634 the quantity a cortical wave measurement leads with. And $|z|$ is a normal-
635 form radius, which a measured envelope is not; this work identifies the two
636 nowhere. Supplemental Material gives the sweep, its uniform control, the
637 amplitude layer, the standing wave it admits inside the band, the potential's
638 curvature at a winding and the defect theorems, together with the remaining
639 limits of both.

640 A vanishing resultant is the absence of global agreement and not of agree-
641 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a dis-
642 crepancy bound of $2\sqrt{2}L|\sin\phi|$, with ϕ the phase advanced across the patch,
643 which falls to zero as the same winding is spread over more sites (`chord_le_`
644 `of_char_patch`), while the global form of the same bound is vacuous at $r = 0$.
645 Encoders shared on overlaps therefore still bound content disagreement in-
646 side a patch of a state that no global observable calls coherent. The winding

647 number itself stays unconnected to content. It is, however, a property of the
 648 state rather than of the cover read on it. On a ring the whole turns dropped
 649 by the winding's affine local lifts are zero at every site but the wrap, where
 650 the residue resets, and they sum around the loop to q ; a nonzero q obstructs
 651 every global real-valued phase lift, so one state at one coupling has resultant
 652 exactly zero and degree q (`winding_degree_obstructs`). Reweighting the
 653 sites moves the first quantity and redrawing the loop's intermediate transi-
 654 tions leaves the second where it was. The scope is the ring's: this is a loop's
 655 winding and not a spiral, the transitions are those of the affine lifts rather
 656 than of an unwrapped measurement, and the integer is identified with no
 657 cortical observable.

658 This qualifies what a phase trace can establish. An onset criterion stated
 659 in r alone registers a change of basin as readily as a change of coupling.
 660 The protocol of Section 8.3 therefore needs a spatial observable beside the
 661 resultant.

662 **7.4 Coherent adaptation can lose environmental align-** 663 **ment**

664 An adaptive coupling rule might supply representational structure, but its
 665 objective must reward the relevant agreement. For symmetric coupling, pair-
 666 wise sine terms cancel in the population mean drift. Minimizing squared drift
 667 at that fixed mean therefore rewards equalizing drift across nodes. In the

frequency-cluster control, this favors coupling between unlike-frequency clusters, whereas the diagnostic environmental template rewards coupling within clusters.

The formalization proves descent of the squared-drift objective for a specified gradient flow at fixed phases. Its functional differs from thermodynamic entropy production: at a coherent stationary density the probability current vanishes while phase-averaged squared drift is positive. Supplemental Material gives the operator, its gradient and the conditions of the descent result.

The finite control co-evolves noisy phases and symmetric, nonnegative couplings at fixed total resource. Its template is a diagnostic covariance of latent signals, never a training input. Across three seeds, second-half order remains between 0.9167 and 0.9222, while the within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel controls are reported in Supplemental Material.

The immediate post-update objective reduction disappears when averaged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its par-

691 ticular objective. Learning models can explicitly include data in an objective
 692 (Corberi et al., 2026); an account of agreement about the cup must specify
 693 the content its learning rule is meant to preserve.

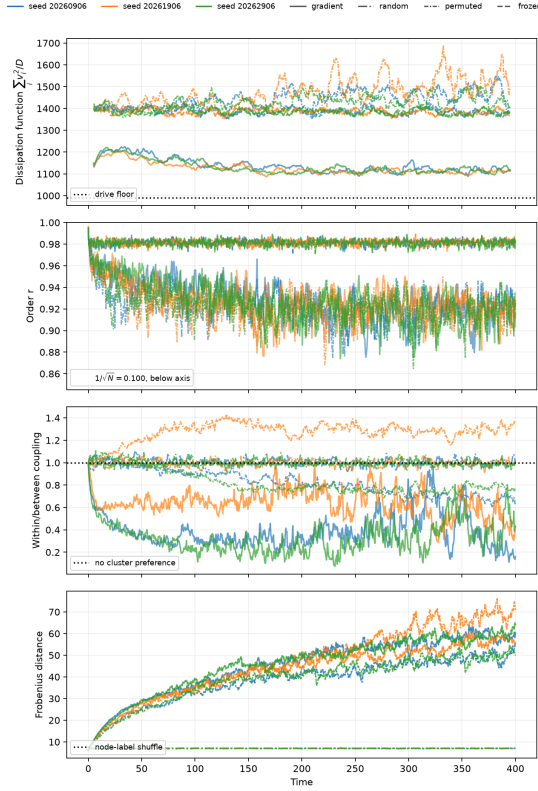


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

694 7.5 Content agreement needs independent decoders

695 A neural test needs independently identified domains, local descriptions and
696 restriction maps. Two decoders should estimate a shared content on over-
697 lapping domains before their agreement is examined. Discrepancy requires
698 a null under decoding noise and controls using deliberately compatible and
699 incompatible examples. An implementation on constructed data separates
700 these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by
701 trial; the compatible construction scores 0.476 on an interval containing one
702 half. Shared-prior shrinkage can nevertheless manufacture agreement, and
703 regional bias can manufacture disagreement. Per-patch decoding diagnostics
704 detect these failures; the compatibility statistic alone does not.

705 Phase agreement cannot replace that measurement. On constructed con-
706 tent, phase carries a share 1.000 of the content entropy when content is built
707 from it and 0.000 when it is not. A phase-derived statistic reports com-
708 patibility on phase-locked patches whose local descriptions disagree. The
709 implemented content statistic detects their incompatibility. These controls
710 validate a measurement on known constructions; a neural content model and
711 decoder remain to be established.

712 Choosing the cover also chooses a candidate subject: trivial descriptions
713 can agree while carrying little content, and a partition can conceal disagree-
714 ment. The useful choice must therefore be constrained by what neural ac-
715 tivity encodes. The exact composition assumes compatibility of its supplied
716 cover. The content mechanisms above identify additional hypotheses under

717 which agreement can arise; they leave the empirical choice of content and
718 the connection to that cover explicit.

719 **7.6 What an informative phase-density measurement** 720 **requires**

721 The concentration range and the dependence structure decide whether the
722 concentration-coherence relation can be tested at all. Under the specified
723 log-density estimator and bias-floor criterion, discrimination at the observed
724 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
725 designs, rising to 31000 when sites share a mean in groups of 10. These
726 counts describe that estimator and dependence model; the intended recording
727 requires its own calibration. Supplemental Material reports the bin-count
728 controls and the full dependence-aware design.

729 That ceiling comes from exercising the estimators on public propofol-
730 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms windows.
731 Bipolar and common-reference montages produce different residuals. The
732 bipolar results are compatible with the Bessel relation over a concentration
733 range too narrow to distinguish it from its linear approximation at the ob-
734 served scatter, which is the range the design scan takes as given. Temporal
735 pooling changes the estimand, and sedation recordings leave the awakening
736 hypothesis untested.

737 An informative dataset therefore needs sufficient concentration range or

738 precision, calibrated estimators and an observation model for dependence,
739 source mixing and reference choice. Public mouse calcium imaging provides
740 thousands of spatially located neurons across waking, sleep and anesthe-
741 sia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-
742 estimation band used here. Site count, state coverage and bandwidth are
743 separate requirements.

744 **8 The cortical hypothesis and an awakening**

745 **test**

746 The cortical proposal identifies a physical source of coupling and asks whether
747 its changes help explain recovery. The tests in Section 7 determine what
748 observations must distinguish before a phase trace can support this identifi-
749 cation.

750 **8.1 Physical identification and calibration**

751 The cortical hypothesis identifies part of effective coupling with endogenous
752 extracellular electromagnetic interactions. Fields can modulate spike timing
753 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
754 propose coordination of distributed neural information (Pinotsis & Miller,
755 2022; McFadden, 2020). Phase organization under propofol motivates study-
756 ing this regime (Bastos et al., 2021), but does not measure coupling.

757 The formal field predicate requires a jointly continuous kernel, a normal-
758 ized substrate, coupling equal to the kernel average, and no single site pro-
759 viding more than half that strength. Toy constructions satisfy it. A cortical
760 realization must specify the scale at which the kernel and its reduction ap-
761 ply: calcium imaging reports different state-dependent modularity patterns
762 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

763 Cortical columns are the obvious candidate unit, and they answer one
764 of the two demands the formalism makes. As the oscillator index they are
765 a serious proposal: they are delineated by response properties and anatomy
766 rather than by the agreement they would be used to establish (Mountcastle,
767 1997), and a phase can be estimated from a column-sized territory at the
768 electrode spacings used for cortical wave measurements (Townsend et al.,
769 2015). As the content cover they are a poor one, for three separate reasons.
770 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
771 the compatibility obligation without constraining anything (Section 4). The
772 cover condition additionally requires the patches to exhaust the substrate,
773 while much of association cortex has no accepted columnar parcellation. And
774 the column’s standing as a canonical cortical unit is contested (Horton &
775 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
776 supplies the oscillator index and leaves the content cover to be identified on
777 separate evidence, which is the bridge assumption E78 records.

778 The spatial reduction is also testable. A row-sum aggregation at zero
779 diagonal tracks the uniform-coupling control over the tested spatial decay

780 range, while frequency heterogeneity displaces the operational threshold. The
781 fit window biases both estimates relative to $2D$; agreement with the control
782 is therefore the supported conclusion. Supplemental Material reports the
783 reduction, its numerical error and the distinct assumptions needed for the
784 kernel-average identification in Eq. (5).

785 The field commitment requires physical evidence beyond a choice of math-
786 ematical representation, and the available support comparisons do not supply
787 it. A finite matrix spread over positive-mass cells preserves its total cou-
788 pling and phase correlation exactly, while point-supported vanishing under
789 an atomless measure and a disadvantage from excluding a better-correlated
790 pair concern specified support constraints; Section 9.3 reads the same com-
791 parisons against a digital candidate. Neither supplies energy prices.

792 A spike-time shift alone supplies no coupling rate. The dimensionless
793 product of population size N , timing sensitivity s , field amplitude E and
794 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

795 This rate describes how induced timing shifts accumulate into drift. Nei-
796 ther it nor the installation factor κ follows from a heat bound. Testing the
797 physical threshold requires independent estimates of the interaction kernel,
798 its aggregation and diffusion in the same cortical regime.

799 8.2 Extracellular geometry and recovery

800 The proposed control variable is extracellular geometry. Extracellular vol-
801 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
802 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
803 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
804 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
805 an effective-medium description predicts that a larger extracellular fraction
806 raises conductivity and reduces field amplitude.

807 The coupling-gated hypothesis consequently predicts that extracellular
808 contraction on awakening raises E and, holding the other factors in Eq. (15)
809 fixed, raises effective coupling. This argument supplies a direction; the
810 geometry–coupling relation, its magnitude and its role in human recovery
811 require measurement.

812 For a smooth crossing at nonzero speed, evaluating the stationary branch
813 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

814 This follows from Eq. (3) at fixed D and concerns the stationary branch
815 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
816 dently justified tracking approximation. The crossing controls in Section 7.2
817 leave that approximation unsupported at onset, even though a separation of
818 geometry and phase time scales can support it away from threshold. The

819 awakening hypothesis therefore needs a calibrated dynamical prediction for
820 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
821 does not supply a recovery latency.

822 **8.3 A protocol for the awakening experiment**

823 The physical test has six requirements:

- 824 **1. Coupling and diffusion in the same units.** Estimate them from re-
825 solved phase dynamics, separating heterogeneity and observation noise.
826 Measure the drive or perturb the system to distinguish endogenous cou-
827 pling.
- 828 **2. Geometry calibrated to coupling.** Validate the geometry observ-
829 able and its relation to effective coupling, with uncertainty. The con-
830 ductivity argument supplies only the expected direction.
- 831 **3. Calibrated phase measurements.** Estimate r by resultant length
832 and a by unweighted regression of log density on centered cosine.
833 Choose site count and binning from a dependence-aware calibration
834 over the intended concentration range. The tested designs require
835 thousands of sites at the observed concentrations; temporal pooling
836 changes the spatial estimand.
- 837 **4. An independent recovery endpoint.** Measure behavioral recovery
838 separately from phase organization.

839 **5. A predeclared dynamical onset criterion.** Set a coherence thresh-
840 old above the finite-size floor and evaluate its latency under the mea-
841 sured coupling trajectory, including ramp speed or fluctuations. Record
842 a spatial observable beside the resultant, since a winding field reports
843 a low resultant at coupling that sustains local order (Section 7.3).

844 **6. Matched model comparison.** Compare coupling increase, diffusion
845 change, common drive and alternative recovery models under the same
846 observation and filtering assumptions.

847 Failure of the concentration–coherence relation over an informative range
848 constrains the density model. Failure of the calibrated geometry–coupling
849 association constrains the physical gate. Better prediction of independent
850 recovery by a competing mechanism counts against its proposed role. A
851 favorable outcome still leaves the content of coordinated activity to be ex-
852 plained.

853 **9 Discussion**

854 Two things have been established here, and they are of different kinds. The
855 constraints of Section 2 hold outright: synchrony does not deliver compatible
856 content, the resultant does not determine the state, a unique fixed point is
857 not a reconstruction, a region that cannot resolve a family cannot encode it,
858 and agreement about a change takes communication. Any account that reads
859 unity off phase coherence has to meet them, this one included. The account

860 set out in Section 1 is conditional at every step by contrast: each connection
861 in it is a hypothesis supplied rather than a result derived, and Table 1 names
862 all eight. Its four non-standard ingredients, named in Section 1, are the
863 installed-energy condition, the patch-local form of the content bound, the
864 criterion separating a winding field from weak coupling, and the machine-
865 checked composition with its joint witness. What follows takes the conditions
866 in turn, as commitments the proposal incurs, as measurements that would
867 bear on them, and as questions the framework leaves open.

868 **9.1 Unity and a minimal self as empirical commit-** 869 **ments**

870 We propose the state glued from compatible local descriptions as a corre-
871 late of conscious unity, and its accurate internal reconstruction as a correlate
872 of a minimal self. The motivations are that different contents belong to
873 one experienced situation and that the situation is experienced from a per-
874 spective. The conditional results identify relations a realization would need:
875 sufficient installed resources for the specified phase model, agreement about
876 shared content, and an encoding that supports reconstruction at the required
877 fidelity. Discharging E12–E89 for cortex would make the composition a state-
878 ment about cortex: that a cortical field above threshold relaxes to a cover
879 whose compatible local descriptions glue, and that the glued state is the
880 unique fixed point of a reconstruction map realized there. It would leave the

881 two identifications above exactly where they are, because neither is among
882 the eight and neither follows from their conjunction; they are the commit-
883 ment this subsection names, and Section 9.2 lists the three observations that
884 would count against it.

885 Compatibility allows different descriptions to contribute different infor-
886 mation. It also permits uncertainty and conflicting judgments: apparent cup
887 position and judged cup position can be distinct quantities in a consistent
888 description of their discrepancy. Which quantities are shared must be de-
889 termined independently. Choosing them solely to secure agreement would
890 deprive the proposal of empirical force.

891 The minimal-self proposal concerns the perceiver’s position and condition
892 within the situation, such as the cup’s location relative to a body that can
893 reach it. The encoding could be implicit and need carry no autobiographi-
894 cal narrative. Its proposed necessity is philosophical: even an unreflective
895 episode has a perspective. Gluing by itself requires no self-encoding, and re-
896 constructability alone supplies no functional role. Altering the encoded hand
897 or viewpoint while holding the external scene fixed should change estimates
898 of the perceiver’s relation to the cup as the readout predicts. Whether that
899 represented perspective is experienced remains a further question.

900 **9.2 What a test of the full proposal must establish**

901 Both uniqueness results concern specified objects. They choose neither the
902 cover nor the encoding, and do not select between candidate subjects. In-

903 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
904 here selection requires independent evidence about neural organization. A
905 constant readout can contract and fix a predetermined state while discard-
906 ing its input. The family criterion in Section 5.2 tests the sensitivity that
907 this construction lacks, provided the relevant states and fidelity are declared
908 independently.

909 A test must separately measure coupling relative to diffusion, content
910 compatibility and self-reconstruction, alongside an independent endpoint for
911 experience. Meeting all three without a unified episode would challenge suffi-
912 ciency. A unified episode with incompatible descriptions would challenge the
913 unity identification; one lacking the specified self-encoding would challenge
914 the reflexive requirement. Awakening, dreaming and dissociative states offer
915 possible settings, but absence of report cannot by itself define absence of
916 experience.

917 The next physical requirement is a calibrated geometry–coupling relation
918 and resolved phase dynamics. The installation bound contributes a nec-
919 essary resource condition where its hardware representation applies; cortical
920 mode prices and field-mediated coupling require measurement. For the repre-
921 sentational connection, an informative experiment asks whether a controlled
922 change in coordination alters independently decoded content agreement, with
923 controls for sensory information and decoder quality. The constructed-data
924 estimator establishes feasibility for specified contents. Neural local descrip-
925 tions, shared overlap encoders and the self-encoding have yet to be identified.

926 The population-size dependence of the phase-to-content bound further limits
927 its quantitative use, in the form Section 4.2 reads off the sheet.

928 The separation results explain why these measurements must be inde-
929 pendent. A stationary distribution leaves several recovery mechanisms indis-
930 tinguishable. Equal phase order can accompany incompatible descriptions
931 or different task information. The tested plasticity rule preserves coherence
932 while losing its designated template. Each finding concerns the specified
933 model and observable; their role is to delimit the evidence needed for the
934 proposed connections.

935 **9.3 What the conditions constrain in a digital candi-** 936 **date**

937 The results of Section 2 hold for any candidate, and for a large language
938 model (LLM) running on graphics processing units (GPUs) four of them are
939 checkable rather than merely stateable. Those four are where this framework
940 bears on claims made for such systems. It does not exclude them, and no re-
941 sult here could; what it does is fix what a claim of unity or self-representation
942 commits its maker to measuring.

943 **Codes, not reports.** Reconstruction across a family is a counting con-
944 straint before it is anything else. If two macrostates the candidate is said
945 to represent satisfy $E(s) = E(t)$ while $d(s, t) > 2\varepsilon$, Eq. (10) rules out re-
946 constructing both within ε , whatever the readout. A system's reports are

947 readouts. Identical verbal reports across two conditions are therefore consis-
948 tent with identical internal codes and settle nothing about the states, while
949 the converse claim — that the system represents its own condition across a
950 declared family — asserts that its internal code separates that family, which
951 is a statement about activations under intervention and not about text. The
952 measurement follows from the statement: declare the family and the fidelity
953 first, then test whether the encoding region distinguishes its members.

954 **The region, not the readout.** Where the encoding is read as the state’s
955 restriction to a region of the machine, the demand falls on the region before
956 any readout exists. For a decoder-only transformer the causal mask makes
957 the network an instance of the update model of Section 5: the activation at
958 position t and layer ℓ reads positions at most t at layer $\ell - 1$ and nothing
959 else, so a region’s causal past is fixed by the architecture rather than chosen
960 by interpretation. The deadline bound of Section 5 therefore applies with
961 its communication graph given rather than assumed. A claim that a model
962 reports on its own processing consequently carries an obligation to say which
963 positions and layers the report’s causal past contains, and a report that
964 appears to track an intervention outside it indicates a confound or a different
965 region, not a faster channel.

966 **Agreement about what.** Unity requires the candidate’s local descriptions
967 to agree on independently identified shared quantities (Section 4). Common
968 timing, a shared residual stream and fluent output supply none of that, for

969 the reason Section 2 gives: coordination bounds content disagreement only
970 through encoders already related on overlaps, and which quantities two parts
971 of a model describe in common is a decoding question carrying the failure
972 modes of Section 7. The cover is a choice, and choosing it to secure agreement
973 empties the claim.

974 **What is not excluded.** Digital representations can satisfy compatibility,
975 so finite hardware is compatible with the gluing construction and none of
976 the above is a no-go for machines. The coupling-gated route E56–E78 is a
977 different matter: it requires a calibrated physical coupling, its noise regime
978 and a connection from coordination to content agreement, and an attention
979 matrix or a shared clock supplies none of these identifications. Comput-
980 ing a field model likewise leaves its realization in the computing substrate
981 to be established. Those physical obligations fall on the cortical proposal
982 identically.

983 The installed-energy condition of Section 3.2 is restricted in the same
984 way. It constrains a macroscopic candidate only through κ , which is hard-
985 ware data supplied by mode profiles and prices, and Supplemental Material
986 reads the condition against a cortical metabolic budget to return the conver-
987 sion factor it would require rather than a verdict. The hardware comparisons
988 there have similarly restricted scope: fixed wiring can impose a correlation
989 disadvantage, and point-supported kernels vanish under an atomless contin-
990 uum measure, but neither comparison makes the representation conditions

991 unattainable on GPUs. A universal exclusion would need a necessary condi-
992 tion for consciousness and a proof that every relevant implementation violates
993 it, and the present composition supplies neither.

994 Whether experience requires a living substrate is the explicit subject of
995 biological naturalism (Seth, 2026a,b). On that view the properties ground-
996 ing experience are those of self-sustaining organisms, and simulating them
997 does not secure them. The conditions stated here name no biological mech-
998 anism, but computation described as such does not satisfy them either: a
999 candidate must exhibit a physically realized coupling in a measured noise
1000 regime, agreement about independently identified shared quantities, and an
1001 internal reconstruction meeting a declared fidelity. Whether a realization
1002 of those conditions that is not alive would be accompanied by experience
1003 is the point at issue, and these results do not settle it. A related position
1004 locates the difference in the mode of computation rather than in life as such:
1005 biological systems compute across scales that do not separate, and along-
1006 side discrete operations carry continuous-valued dynamics belonging to their
1007 physical medium (Milinkovic & Aru, 2026). The coupling condition stated
1008 here is one measurable instance of that requirement, fixed to a physical field
1009 in a declared noise regime, and a description of such a field does not satisfy
1010 it.

1011 The neuroscientific case against *human-like* consciousness in text-only
1012 LLMs concerns embodied sensory interaction, thalamocortical organization
1013 and biological regulation linking perception and action to survival (Aru et al.,

2023). Their absence supports skepticism, conditional on their causal relevance to experience. Those mechanisms are not requirements of the abstract gluing and reconstruction definitions, and workspace-like representations reported in some LLMs (Gurnee et al., 2026) supply candidate organization to investigate. What this framework adds to that investigation is an order for the questions: declare the family of states and the fidelity, fix the encoding region and read its causal past off the architecture, and only then ask what the system reconstructs. A claim that survives those three steps is a claim about the machine; one that does not survive them was a claim about its output.

9.4 Explanatory scope

Integrated-information, global-workspace and predictive-processing accounts organize unity, access and modelling differently (Tononi et al., 2016; Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a candidate physical interaction and conditional relations between local content and self-encoding. Gluing and reconstruction permit multiple substrates; the electromagnetic commitment enters through physical calibration. The formalization checks implications under their hypotheses, and its witnesses establish realizability on declared models. Two things stand outside that result and are not supplied by it: the eight hypotheses are discharged for toy components rather than for cortex, and the identification of the glued state with unity and of its reconstruction with a minimal self is a further

1036 commitment that the eight do not contain.

1037 The explanatory question is whether the specified relations occur in neural
1038 activity and matter for experience. In the cup example, this means establish-
1039 ing how coordination helps maintain one represented situation and how the
1040 perceiver’s place within it contributes to the episode. The proposal identi-
1041 fies candidate correlates and tests of that identification. Why those relations
1042 would be accompanied by experience remains unexplained.

1043 **Data and code availability**

1044 The Lean development, simulation scripts, tests and saved numerical sum-
1045 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
1046 `cortical_coherence`. Publication simulation values are generated from
1047 saved summaries without rerunning production sweeps. The exploratory
1048 EEG analysis uses OpenNeuro dataset `ds005620` (Bajwa et al., 2025); pre-
1049 processing and selection are described in Supplemental Material. The repos-
1050 itory contains the reproducible analysis and does not redistribute the raw
1051 recordings.

1052 **Funding and competing interests**

1053 This research received no external funding. The author declares no compet-
1054 ing interests.

1055 Use of AI tools

1056 AI assistance to this work was substantial and is not separable by task. Sev-
1057 eral coding assistants and chat interfaces were used in an interleaved way
1058 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
1059 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
1060 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
1061 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
1062 accessed with the Hermes CLI. Between them they contributed to the Lean
1063 development, the simulation and analysis code, and the drafting and revision
1064 of both documents. Because the tools were used in combination and in alter-
1065 nation, no section, proof, figure or result is attributable to any one of them,
1066 and none is claimed to be. The author is responsible for all content, including
1067 everything a tool produced. Machine checking establishes the stated Lean
1068 results under their hypotheses; it does not validate biological interpretations
1069 or replace scientific review of generated code and prose.

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